Final Strategic Report: Olist Store Performance (2017 - 2018)

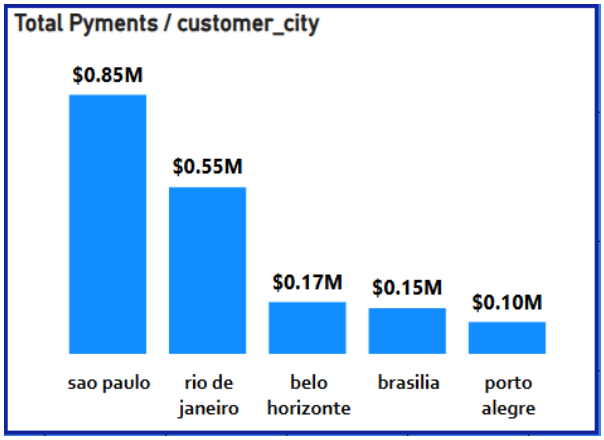
First: General Store Performance Report

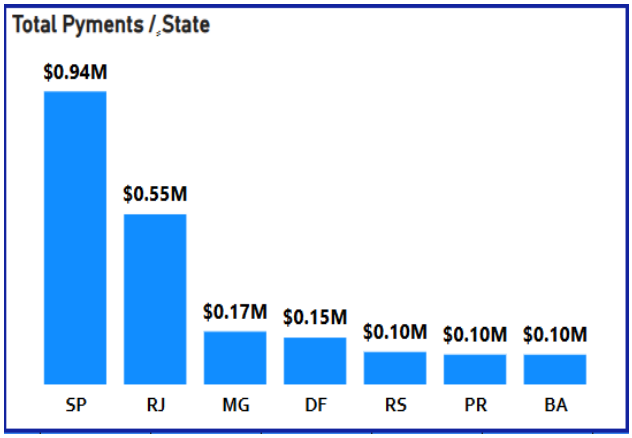
Analysis of the highest-revenue cities:

- **São Paulo** is the primary source of revenue, significantly outperforming other cities in total payments (by approximately US$1 million), reaching around US $2.11 million. This demonstrates its strategic importance.

- **Rio de Janeiro** contributes roughly half of **São Paulo's** revenue, while the revenue of other cities does not exceed US$1 million.

- purchasing is concentrated in **São Paulo** and **Rio de Janeiro** (the capitals of the states of SP and RJ), as these are the most vibrant cities and the primary sources of store revenue.





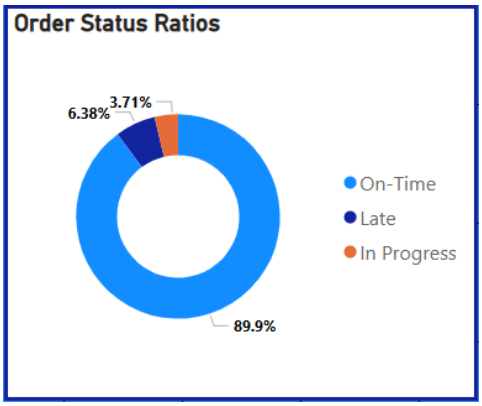
Final Insights:

- This indicates that the store relies on only two main revenue streams.

- This poses a strategic risk to revenue sustainability, as any issues related to purchase or supply prices for branches or other factors in these two cities will have a significant impact on the e-commerce store's financial viability.

Delivery situation analysis:

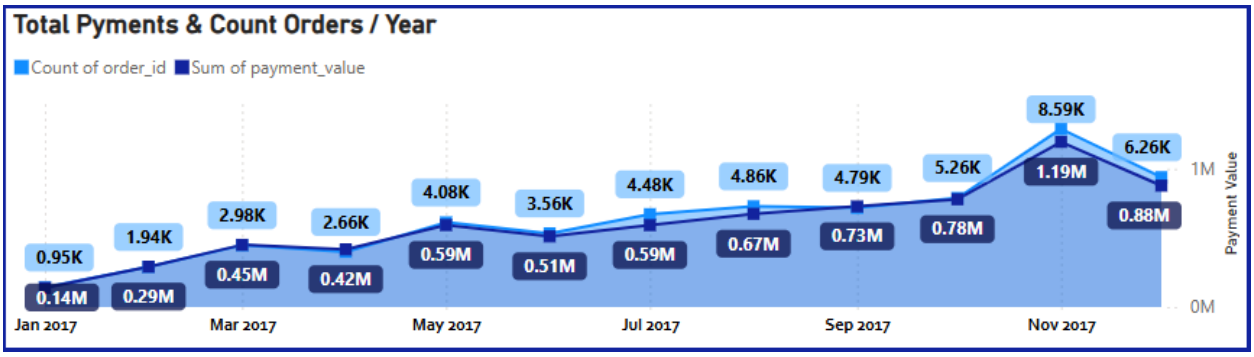
The percentage of orders delayed beyond their scheduled delivery time is approximately 8%, a high rate compared to other stores, as around 9,000 orders experienced delays. This high percentage negatively impacts the store's reputation and overall rating.



Time-based analysis of payments:

The timeline for total sales and payments shows a significant increase from the beginning of 2017 through January 2018, with an increase of approximately 10,000 orders and $1.4 million.

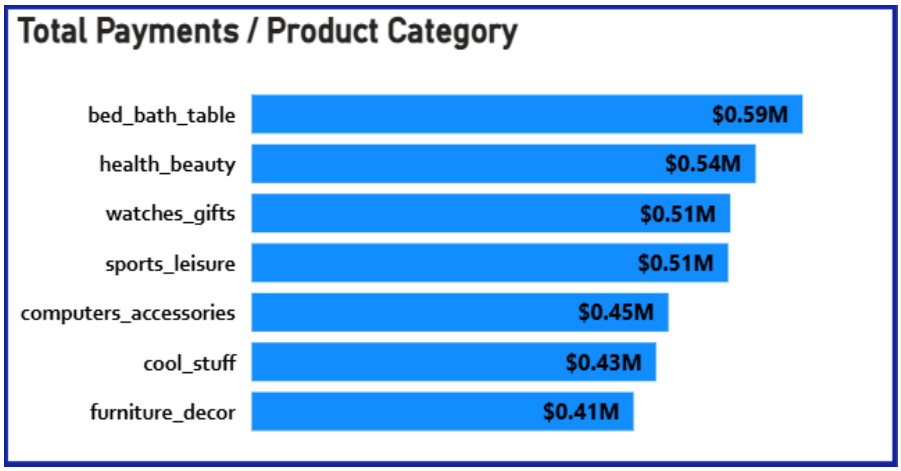
Note: The order data covers the period from January 1, 2017, to August 31, 2018. The data has been filtered to represent the middle of these two periods. For details on how to adjust the data, please refer to the data cleaning steps page.



Analysis of the highest-earning product categories:

- The health_beauty category leads in revenue with approximately $1.4 million and a total of 9,552 orders.

- Watches_gifts follows closely behind with about $0.14 million less and a total of 5,644 orders. Bed_bath_table comes in at a very slight difference in revenue with 11,015 orders. The remaining categories on the list see a gradual and almost equal decline in revenue from one category to the next.



Insights:

- No single category dominates revenue; it can be argued that seven product categories contribute significantly to the store's income, indicating a diverse product portfolio and potentially high reliance on a single product.

  Note: This is a general overview. Detailed analysis can be found on the Payment Analysis page

- The bed_bath_table category stands in contrast. Its third-place ranking is due to high demand. Total demand is almost double that of the watches_gifts category, yet it still outperforms it, suggesting that its products offer a compelling buying opportunity with a low profit margin.

Second: Logistics & Delivery Report:

Analysis of delivery and shipping days for the cities with the most delivery days:

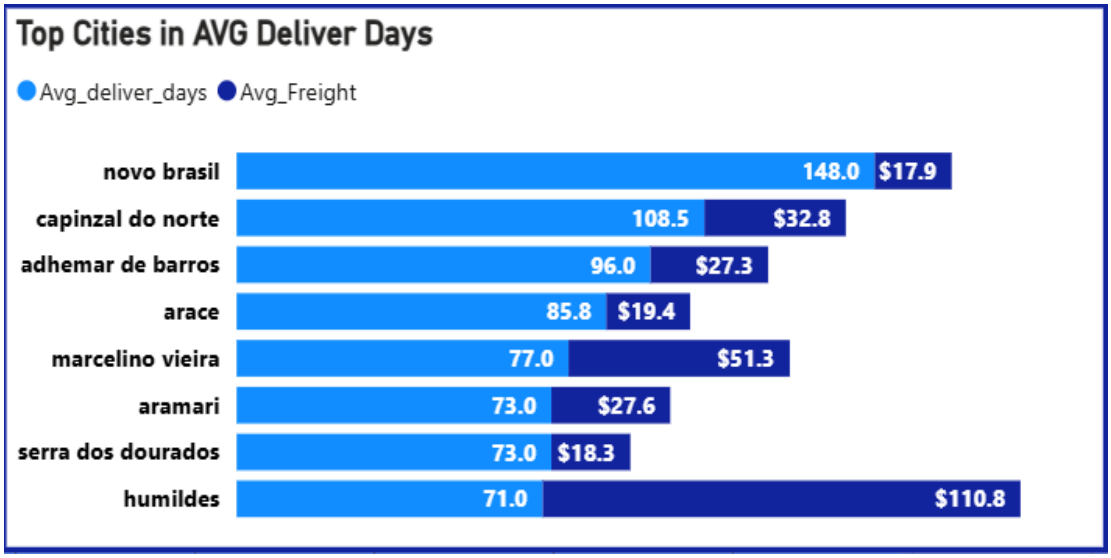
- Novo Brasil leads the list of cities with the highest average delivery time, 40 days ahead of the next highest, and boasts the lowest average shipping cost ($17.9). Its distance from the store's logistics network places it in a moderate position.

- Humildes has the lowest average shipping cost at $110.8 and is located in a remote position.

- Data indicates that the top eight cities (Novo Brasil, Capinzal do Norte, Adhemar de Barros, Arace, and others) have average shipping costs under $35, with the exception of Marcelino Vieira and Humildes.

- Arace, Aramari, and Capinzal do Norte are far from the logistics network, yet their average shipping cost remains below $35, a low figure comparable to the average shipping cost of nearby cities like Adhemar de Barros. Even in the two capital cities, there was a 7.6% delay, indicating either supply chain disruptions or a general miscalculation of delivery times.

- Approximately 1.4% of orders, totaling $0.2 million, were found to be non-category items (blank). This category was excluded from the category analysis to ensure accurate comparisons between known product types, while remaining in the store's total sales figures to ensure consistency with the final financial data. Parallel to the scatter plot, the relationship between average shipping cost and average delivery days is inverse; that is, the number of days decreases as shipping cost increases, particularly in the excluded cities.



Insights:

- Countries with lower shipping costs experience longer delivery times, with the exception of some countries where the increased time may be due to distance or other factors.

- Consistent with the parcel failure rate chart, a 1.19% failure rate is indicated. This suggests a failure rate in reaching the precise geographical location in these cities, or that the failure is due to inadequate road networks and infrastructure in those cities.

Recommendation:

- Develop a well-considered plan for a flexible shipping pricing system that takes into account distance, speed, and shipping efficiency.

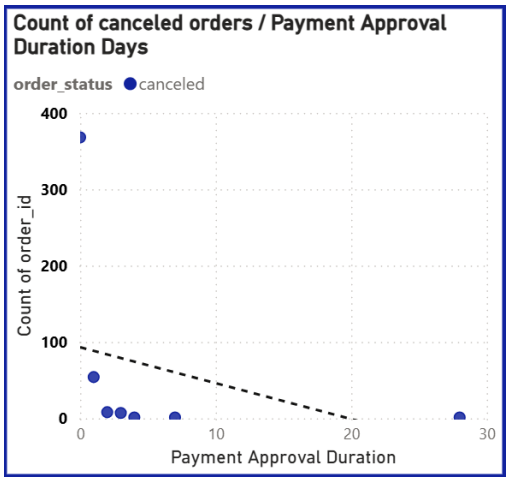
- Contract with local fulfillment centers in northeastern Brazil to reduce travel distances in states far from southeastern Brazil.

- Re-evaluate logistics partnerships in cities with above-average failure rates and seek to contract with more efficient companies.

- Recommendation: Adopt smart address verification technologies, especially in remote cities with weak road networks.

Does the time it takes for bank approval affect the order cancellation rate?

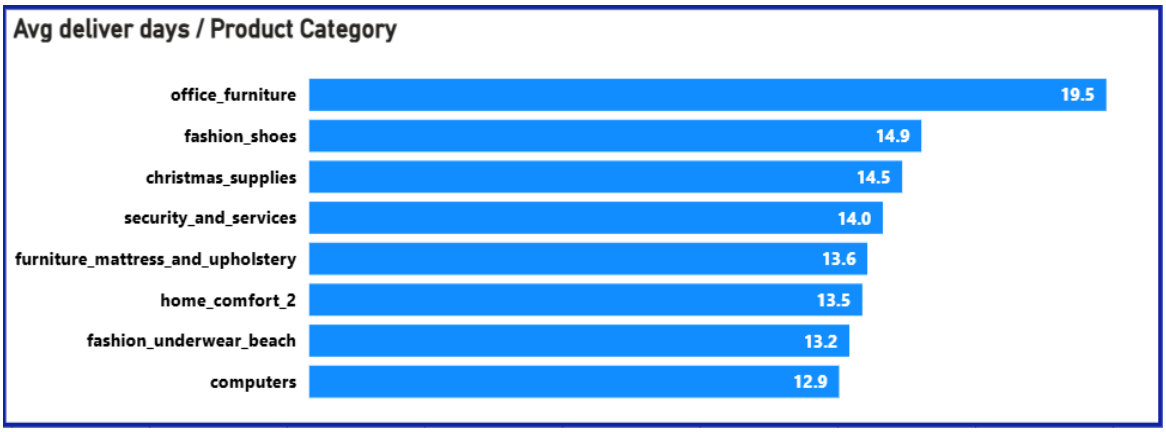
Answer:
The scatter plot analysis showed that the majority of canceled orders occur within one day of bank approval, indicating an inverse relationship. This suggests that delays in bank approval do not directly influence a customer's decision to purchase.



Analysis of delivery days for product categories:

- The data indicates that categories with higher volumes, such as office_future and home_comfort_2, and those sensitive to damage or breakage, such as computers, security_and_services, and Christmas_supplies, have the longest average shipping times, with the exception of fashion_underwear_beach and fashion_shoes.

- Contrasting with the scatter plot between volume and shipping times, a direct relationship is observed between them. This suggests that volume and sensitivity to damage are direct factors affecting shipping times. The exceptions to the categories may be due to supply chain issues or the distance of the supplier.



Recommendation:

- Contract with local distribution warehouses specializing in large and sensitive categories to reduce risks and delivery time.

- Conduct a study to determine the reasons for delays in less demanding categories such as fashion shoes and beachwear to understand the increased delivery times and address the issue.

- There is an inverse relationship between the number of product images available in the store and the number of orders for that product. This indicates

Analysis of the number of images factor on demand rate:

Data indicates that demand for fashion products increases when they have 2, 3, or 4 images. Meanwhile, demand for Home and Healthy products is concentrated on those with a single image. Construction products with a single image are the most popular across all categories. In other categories, the majority of products have a single image.

Recommendation:

Develop a well-thought-out marketing plan that includes providing products with an appropriate number of images, based on demand. This plan should specifically include the following:

1- Mandate that fashion product sellers provide a minimum of 4 professional images to ensure competitiveness.

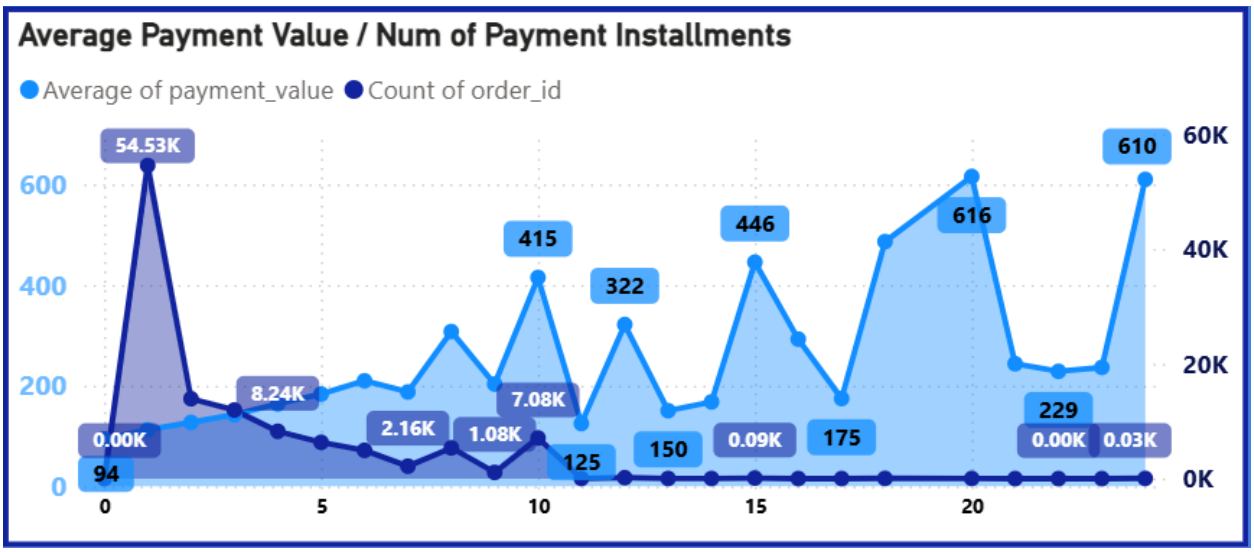
2- Recommend that products with high demand for a single image should have a comprehensive, high-quality image that includes all the details relevant to the buyer.

3- Conduct a pilot program to increase the number of images for products with high demand for a single image but which experience a general decline in demand. This will help determine whether the image factor is a direct influence or if the decline is inherent to the product itself.

Third: Payments Report

Installment analysis in terms of demand and amounts paid

The data indicates a direct correlation between the number of installments and the average payment. Despite the convenience of installment plans, the vast majority of purchases are made with a single, full payment, as the average payment is $112.42. This suggests that the majority of the store's revenue comes from low-cost purchases and a general preference for direct payment over installments.



Recommendation:

1- Clearly highlight the installment payment option next to the price of high-priced products such as electronics and furniture to encourage low- and middle-income customers to purchase these products and increase demand.

2- Incentivize cash payments for smaller items by offering small discounts of no more than 5% for immediate payment to increase the store's cash flow without waiting for installment payments.

Analysis of total amounts, margin, and purchase rate for the top 6 product categories:

- The data indicates that the watchets_gifts and health_beauty categories, despite having the lowest profit margins (12.18% and 16.54% respectively), have the highest total payouts ($1.4M and $1.26M respectively). Therefore, the relatively low profit margin translates into substantial cash value, boosting the overall return on investment.

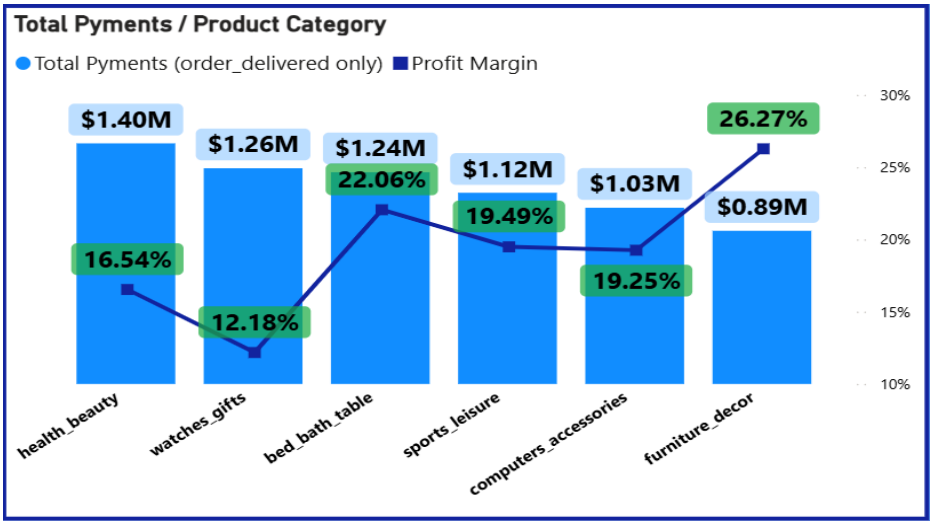
- The watchets_gifts category has the lowest demand of all the categories on the list, yet it competes with health_beauty, making it the largest in terms of net payout. This category is generally considered dynamic and lucrative.

- The bed_bath_table category has a high profit margin of 22.06%, yet it generates approximately the same total payouts as watchets_gifts (around $1.24M). This suggests that while the net payout is small, it is high relative to its cost. The high demand for this category contributes to its overall competitiveness, making it a stable category.

- The computer_accessories and sports_leisure categories represent stable purchasing power, income, and average profit margins, making them well-balanced for the store.

- The furniture_decor category has a high margin of 26.27%. However, the total amount paid out is the lowest at $0.89 million, and demand is average. Therefore, the high profit margin translates to small cash amounts, which doesn't significantly boost the overall return on investment.

- The store's overall revenue is flexible and not dependent on a single category, which greatly reduces the risk of over-reliance. The availability of stable categories like computer_accessories and sports_leisure further enhances this flexibility.



Recommendation:

1- Implement marketing campaigns targeting key categories such as watches_gifts to boost demand and maximize revenue.

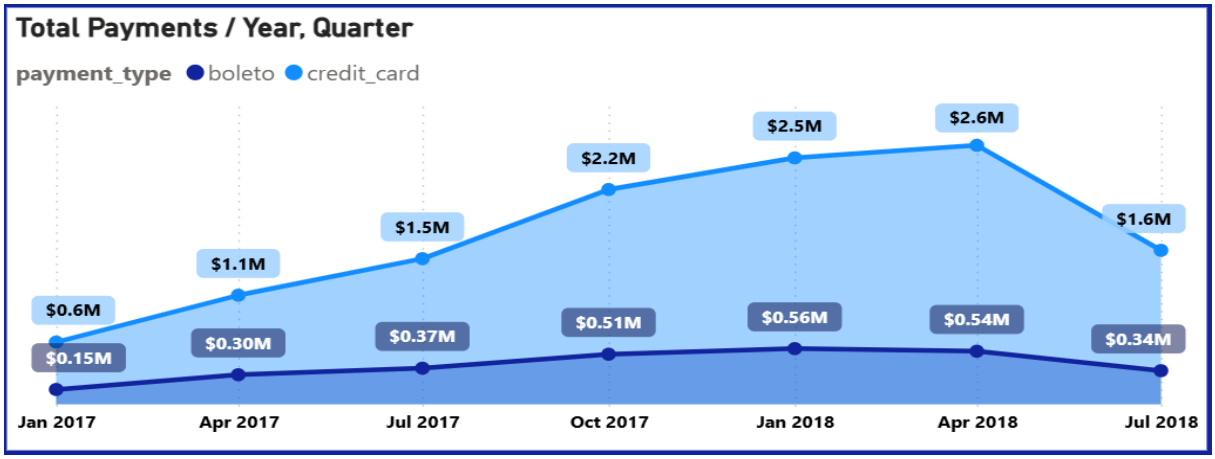
2- Encourage demand for the furniture_decor and bed_bath_table categories by offering small discounts that don't significantly impact the high profit margins of these categories.

3- Offer bed_bath_table products (which are already established) as complementary offers for furniture purchases to encourage furniture_decor purchases and increase the overall value of the purchase.

Payment Method Analysis:

1. Data indicates a gradual increase in total payments from January 2017 to April 2018. This period was
   followed by a noticeable decline until July 2018.
2. Credit cards remain the most popular payment method, accounting for approximately 3.6% to 5.3% of
   total **Boleto** payments. This difference increases over time, indicating a gradual rise in their use as the primary payment method, with Boleto becoming a secondary option. Vouchers are used very sparingly, representing only 6% of total payment for orders.
3. The percentage of orders arriving on time is 2% higher when using credit cards than when using
   Boleto. This could be due to several factors: - The time difference between completing purchases using the two methods: credit cards are processed in seconds, while Boleto takes one to two days. - The higher percentage of expensive products, which necessitates faster packaging and shipping.
4. Credit Card Percentage High interest rates will increase cash flow risks and installment debt.

5. São Paulo leads the largest cities in total payments and total credit card balances with approximately $1.68 million, followed by Redio with nearly half the total.



Recommendation:

1- Study the growth period (January 2017 - April 2018) to identify the reasons for the increase (seasonality, marketing factors, supply chains, etc.). Simultaneously, study the reasons for the decline and develop a well-considered plan to address these reasons while implementing the growth factors that occurred during the previous period.

2- Encourage the use of secondary payment methods (**Boleto** and Vouchers) by offering small discounts, especially to high-margin customers. This will ensure an increase in quick revenue streams and reduce cash flow risks associated with credit card installments.

3- Incentivize customers whose credit card purchases were unsuccessful to use either **Boleto** or Vouchers by offering the small discount mentioned in point 2. This will prevent them from missing out on potential purchase opportunities, particularly in cities with high credit card usage, such as São Paulo and Rio de Janeiro.

4- Improve the **Boleto** processing cycle digitally to reduce waiting times or provide more predictable delivery dates to avoid delays.

Fourth: Customer Satisfaction & Reviews Report

Insights:

- There was a gradual decrease in average reviews from July 2017 to January 2018, indicating a factor negatively impacting the ratings.

- Concurrently, the payment schedule showed an increase in total payments made via credit card starting in July 2017, which continued after the decline until April 2018. This may suggest the following:

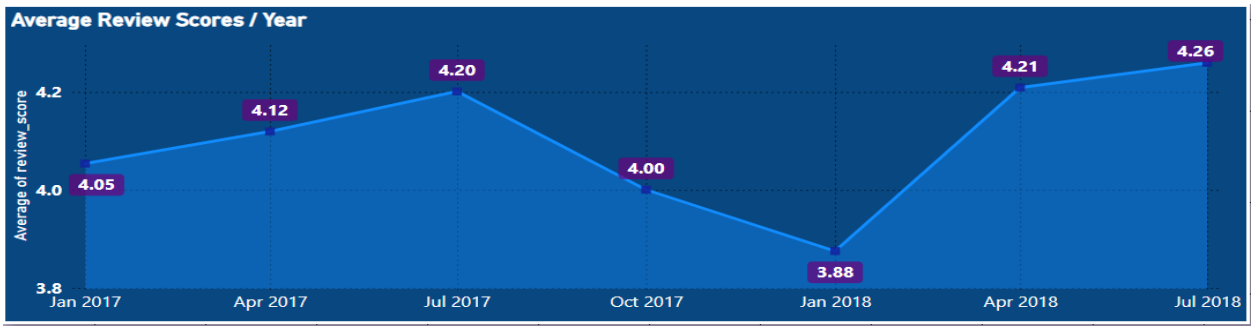
- The reason could be an increase in the average delivery time. An analysis of the delivery schedule reveals a direct correlation between the increase in average delivery time and the decrease in reviews.

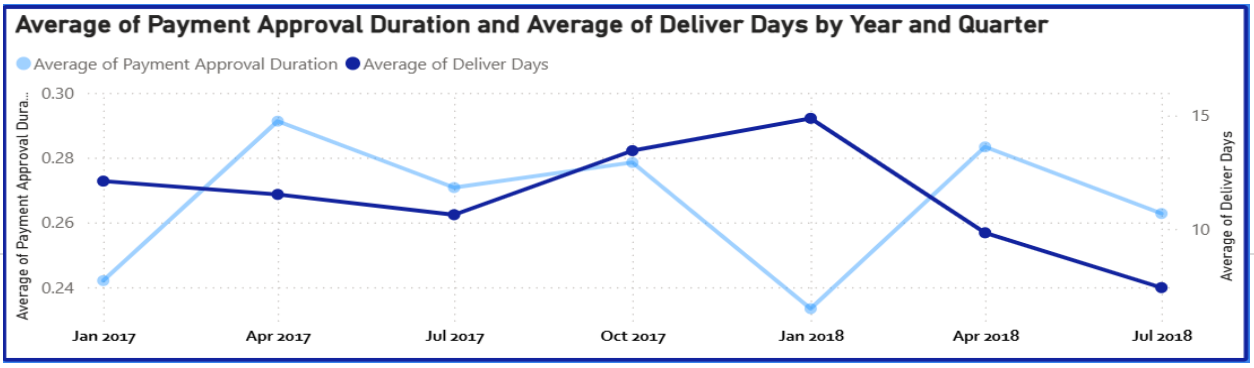
- This confirms that the surge in purchases using this payment method led to a crisis in the shipping and delivery supply chains, resulting in longer delivery times and ultimately a rise in the rate of negative reviews.

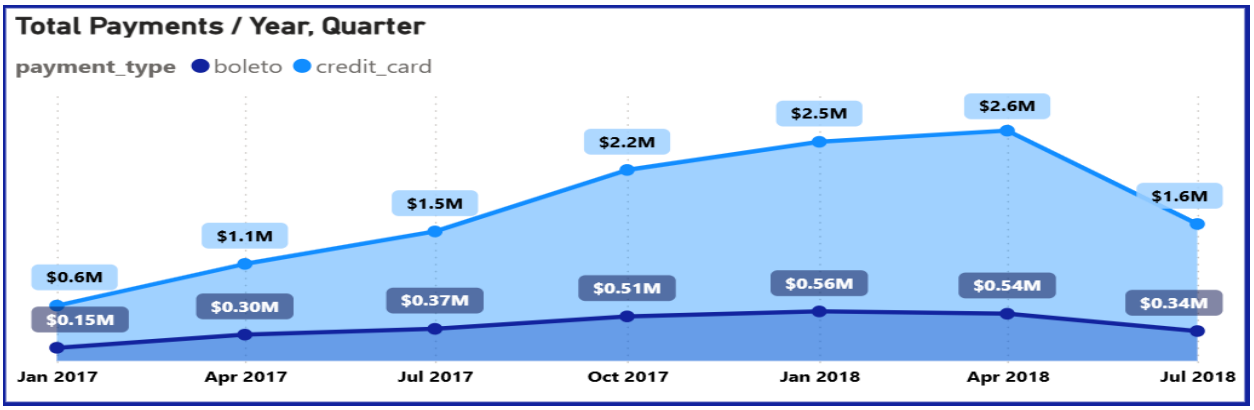
- However, starting in 2018, the problem was addressed. The negative reviews disappeared, and they began to rise significantly, while the total payments continued to increase.

- This suggests a potential corrective measure.

- There is a peak period in the latter half of the year that causes a surge in purchasing activity, particularly with credit cards, indicating the need to prepare for that period by:







Insights & Recommendations:

1. Contract with additional shipping companies during this period.

2. Expedite the processing and packaging cycle to compensate for the long shipping time.

3. Encourage cash payments for small products by offering small discounts of no more than 5% for immediate payment to reduce pressure.

Shipping and delivery time analysis based on the ratings

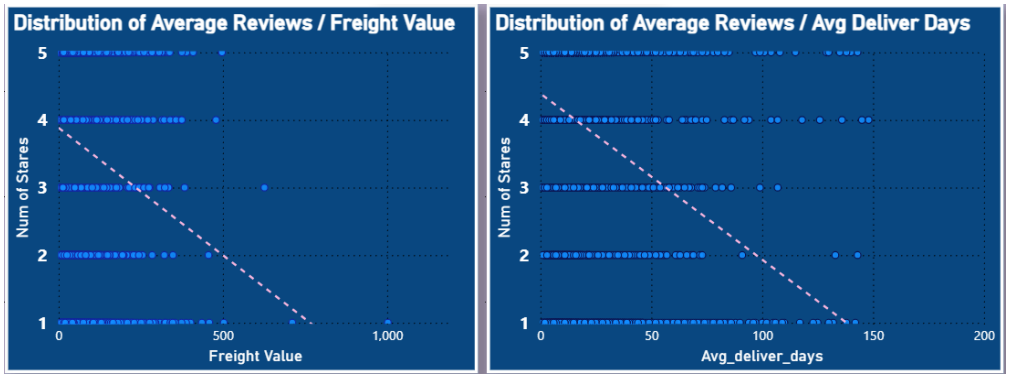
The store boasts strong ratings (+4 stars) across most categories, with the exception of the "Security & Services" category, which suffers from a severe deficiency (less than 3 stars).

There is a clear inverse relationship: the higher the "shipping cost" or "delivery time," the more customer satisfaction plummets, regardless of product quality.

Recommendation:

1- Contract with additional shipping companies and expand operational capacity before July to avoid year-end delays.

2- Immediately review or discontinue the "Security and Services" suppliers to protect the store's overall reputation.



Technical Documentation:

[1] Order_reviews table:

- I deleted older reviews and kept only the most recent ones for the same order by: Sorting the rows with "review_creation_date" and then "review_id", then deleting any rows with the same review_id.

- The most recent review accurately reflects customer satisfaction, and the number of reviews for the same order creates data noise that would hinder analysis.

[2] olist_sellers_dataset table:

- I assigned a code to each seller to display the best sellers by simple name, protecting their privacy and facilitating easy comparison in the "Seller X" graph instead of using a complex ID.

[3] orders_status table:

- I added a column called Final_Status to classify orders as successful or unsuccessful. This is done by examining the order_status column: if the value is delivered, it's successful; if it's delayed or the package hasn't arrived, it's unsuccessful. This helps the calculations later to obtain financial statistics based on successful orders and to get an overall logistics performance evaluation based on the success rate.

- I added a column for the package arrival status: "Arrived on time," "Late," or "In progress." This is done by examining the dates of both order_delivered_customer_date and order_estimated_delivery_date. If they are empty, this indicates that the payment or shipping process has not yet been completed. If the dates are present but the estimated arrival date is earlier than the actual arrival date, this indicates a delay. If the estimated arrival date is earlier than the actual arrival date, then the order is on time. I also added a column to calculate Delivery Days: Understanding Shipping Efficiency and Service Level Agreement (SLA) Compliance

- I noticed an unusually sharp and illogical drop in orders between January 1, 2017, and August 31, 2018. Therefore, I filtered out the incomplete data from that period and retained the data from January 1, 2017, to August 31, 2018, which represents a more typical order volume.

- I added a column to calculate the number of days required for bank approval of a purchase. This allows me to assess the impact of approval on other factors, such as customer reviews and waiting times.